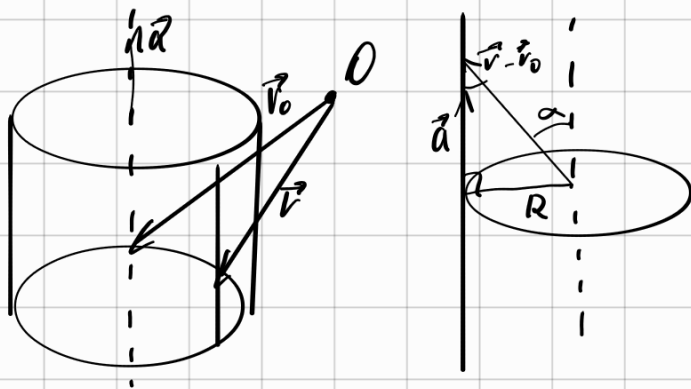


10.26 Локально берем. у-е прямого кругового цилиндра
радиуса R с осью $\vec{r} = \vec{r}_0 + \vec{a}t$



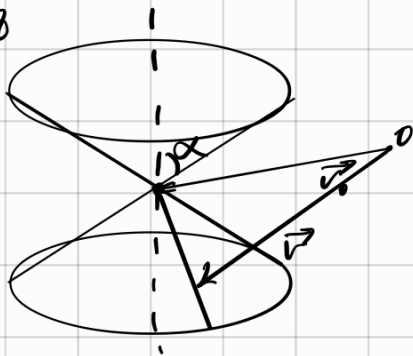
$$\frac{R}{|\vec{r} - \vec{r}_0|} = |\sin \alpha|$$

$$R = |\vec{r} - \vec{r}_0| |\sin \alpha|$$

$$R|\vec{a}| = |\vec{r} - \vec{r}_0| |\vec{a}| |\sin \alpha|$$

$$[\vec{r} - \vec{r}_0; \vec{a}] = R|\vec{a}|$$

10.28



$\alpha(r_0)$ - конуса

$$\vec{r} = \vec{r}_0 + \vec{a}t - \text{ок}$$

α - угол между осью и образующей

$$|[\vec{r} - \vec{r}_0; \vec{a}]| = |\vec{r} - \vec{r}_0| |\vec{a}| |\cos \alpha|$$

10.38 Написать у-е прямого кругового цилиндра, прох ради $\alpha(1;1;2)$ с осью $x=1+t$ $y=2+t$ $z=3+t$

$$\vec{a}(1;1;1) \quad \vec{r}_0(1;2;3) \quad \vec{r}_1(1;1;2) \quad \vec{r}(x;y;z)$$

$$[\vec{r}_1 - \vec{r}_0; \vec{a}] = \begin{vmatrix} i & j & k \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = (0; 1; 1)$$

$$|[\vec{r}_1 - \vec{r}_0; \vec{a}]| = \sqrt{2}$$

$$|\vec{a}| = \sqrt{3}$$

$$\Rightarrow R = \frac{\sqrt{2}}{\sqrt{3}}$$

$$[\vec{r} - \vec{r}_0; \vec{a}] = \begin{vmatrix} i & j & k \\ x-1 & y-2 & z-3 \\ 1 & 1 & 1 \end{vmatrix} = (y-z+1; -x+z-2; x-y+1)$$

$$(y-z+1)^2 + (-x+z-2)^2 + (x-y+1)^2 = \frac{2}{3} \cdot 3 = 2$$

10.39 $a(1;1;1)$ $\vec{r}_0(0;0;0)$ $\alpha = \arccos \frac{1}{\sqrt{3}}$

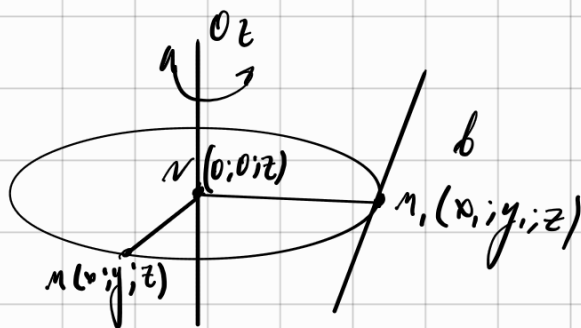
$$\vec{r}(x;y;z)$$

$$(x+y+z)^2 = (x^2+y^2+z^2) \cdot \frac{1}{\sqrt{3}}$$

$$x^2 + y^2 + z^2 + 2xy + 2xz + 2yz = x^2 + y^2 + z^2$$

$$xy + yz + zx = 0$$

10.40 Найми y -е нон. мун ноб-мн баярлан уяран $x=1+t$
 баярлан оон Oz $y=z=3+t$



$$|NM| = |N, N|$$

$$|NM|^2 = |N, N|^2$$



$$x^2 + y^2 + z^2 = x^2 + y^2 + z^2$$

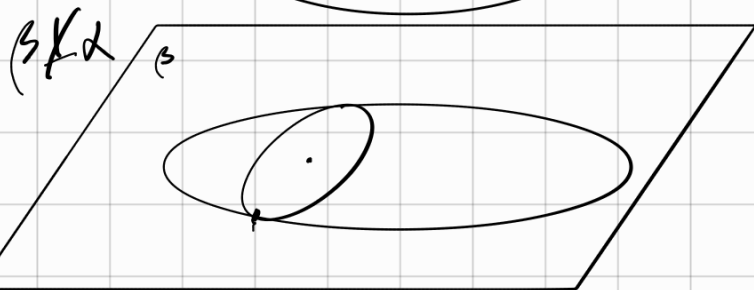
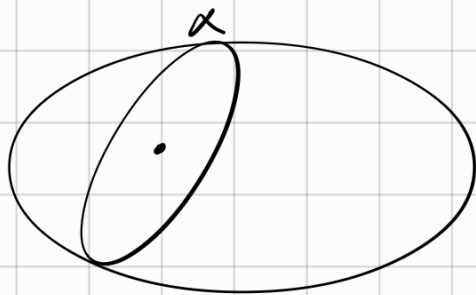
by y -а уяран $y_1 = z$ $x = z$
 $(x_1 = 1+t \quad y_1 = 3+t \quad z = y_1 = 3+t)$

$$x^2 + y^2 = (z-1)^2 + z^2$$

$$x^2 + y^2 = z^2 - 4z + 4 + z^2 = 2(z^2 - 2z + 1) + 2$$

$$x^2 + y^2 - 2(z-1)^2 = 2 \quad ; \quad \frac{x^2}{2} + \frac{y^2}{2} - (z-1)^2 = 1 \Rightarrow \text{огном. мун.}$$

10.65 (2) Найми улам сөрөн $x^2 + 2y^2 + 4z^2 = 40$ $11-10$ $x+y+z=z$



$$x = z - y - z \quad \text{б } y\text{-е } \exists 1-11:$$

$$(z - y - z)^2 + 2y^2 + 4z^2 - 40 = 0$$

$$4y^2 - 14y - 14z + 2yz + y^2 + z^2 + 4z^2 +$$

$$+ 4z^2 - 40 = 0$$

$$3y^2 + 2yz + 5z^2 - 14y - 14z + 4 = 0$$

A 2B C 2D 2E

$$\begin{cases} Ay_0 + Bz_0 + D = 0 \\ By_0 + Cz_0 + E = 0 \end{cases}$$

$$\begin{cases} 3y_0 + z_0 = 7 \\ y_0 + 5z_0 = 7 \\ y_0 = 2 \quad z_0 = 1 \\ x_0 = 4 \end{cases}$$

10.81

Найти уравнение осей.

$$4x^2 - y^2 = 16z$$

выполнить

Ответ: (4; 2; 1)

$$\begin{cases} x = z + a_1 t \\ y = a_2 t \\ z = 1 + a_3 t \end{cases}$$

$$4(z + a_1 t)^2 - (a_2 t)^2 = 16 + 16a_3 t$$

$$4(z + a_1 t)^2 - (a_2 t)^2 - 16(1 + a_3 t) = 0$$

$$16 + 16a_1 t + 4a_1^2 t^2 - a_2^2 t^2 - 16 - 16a_3 t = 0$$

$$(4a_1 - a_2^2) t^2 + 16(a_1 - a_3) t = 0$$

$$4a_1 = a_2^2$$

$$a_1 = a_3$$

$$a_1 = 1$$

$$a_3 = 1$$

$$a_2 = \pm 2$$

$$\text{Ответ: } \begin{cases} x = z + t \\ y = \pm 2t \\ z = 1 + t \end{cases}$$

$$4x^2 - y^2 = 16z$$

$$(2x - y)(2x + y) = 16z$$

$$\textcircled{1} \begin{cases} p(2x - y) = z \\ 2x + y = 16p \end{cases}$$

$$\textcircled{2} \begin{cases} 2x - y = qz \\ q(2x + y) = 16 \end{cases}$$

$$1) \begin{cases} 4p = 1 \\ 4 = 16p \end{cases}$$

$$p = 1/4$$

$$\begin{cases} 2x - y - 4z = 0 \\ 2x + y - 4 = 0 \end{cases}$$

$$\textcircled{2}$$

$$q = 4$$

$$\begin{cases} 2x + y - 4z = 0 \\ 2x - 4y - 16 = 0 \end{cases}$$

